

Preemptive defense against re-identification attacks on voice anonymization via adversarial perturbation

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Abstract—Voice anonymization (VA) is used to conceal the voice identities in speech data. Performance is usually estimated using automatic speaker verification (ASV) as a proxy to judge the capability of an attacker to re-identify speakers after anonymization. The defender anonymizes test utterances; the attacker compares them to equally anonymized reference utterances to infer voice identity, with degraded ASV performance indicating successful anonymization. Thus, the defender’s protection is one-sided and only applied to test utterances via VA. Though unexplored so far, there is an opportunity to enhance anonymization by protecting reference utterances too. We present a new, preemptive VA paradigm: reference utterances are protected using adversarial noise to degrade their potential to infer voice identity. Preemptive protection does not degrade the perceived quality of reference utterances and is independent of the specific ASV system used for identity inference. By combining preemptive protection and regular test-side VA, ASV equal error rates can be increased from 14% to 45% (near-perfect privacy).

Index Terms—voice anonymization, preemptive defense, adversarial noise.

I. INTRODUCTION

Voice Anonymization (VA) [1] is the task of concealing the speaker’s voice identity from a speech signal while preserving other linguistic and paralinguistic attributes. The VoicePrivacy Challenge (VPC) [2]–[4], now in its fourth edition [5], has promoted and fostered progress in the VA research community. In the VPC, the VA task is formulated as a game between a defender and an attacker: given a set of speech utterances, the defender anonymizes them, and the attacker subsequently attempts to recognize the original voice identities.

Early research mostly focused on the defender side [6]–[8], while the attack paradigms experienced a relatively slow evolution. Since the first VPC edition [2], the attack consists in using an Automatic Speaker Verification (ASV) system to identify the speaker within the anonymized utterances, which are used as trials. The attacker compares them to a set of reference utterances of candidate speakers they attempt to re-identify, which are used as enrollment utterances. The higher the Equal Error Rate (EER) of the ASV system, the more difficult it is for the attacker to infer the original speaker’s identity, resulting in a greater degree of privacy protection. Subsequent editions have refined this idea by expanding upon the threat model [3], [4]: namely, if the attacker has knowledge

of what VA system was used to anonymize the trials, they can use it to anonymize their own enrollment utterances and the training data of their ASV system, adapting to that VA system and thus improving it. This remains the standard approach to privacy protection evaluation in the VPC and in the field of VA in general.

More recently, the VoicePrivacy Attacker Challenge [9] has promoted attack-focused research, with novel attack strategies [10], [11] and privacy evaluation metrics [12], [13] being proposed to this day. Notably, [14] introduced the idea that the EER used as an estimate of privacy protection is influenced by two factors: (i) how well the VA system used by the defender can suppress speaker-specific information in the signal, and (ii) how ineffective the ASV system used by the attacker is at detecting residual speaker-specific cues that remain in anonymized utterances. In other words, an estimate of seemingly-strong privacy protection could be due to an *ineffective* attacker ASV system rather than an *effective* defender VA system. The authors name this phenomenon *privacy overestimation* and identify it as a real risk because, in a research setting, the evaluation of a VA system requires the developer to play both the role of the defender and the attacker: if the developer unknowingly uses an ineffective attack, their privacy evaluation will be overly optimistic, resulting in a false sense of protection.

In this paper, we propose a technique that *purposefully* exploits the component of privacy overestimation stemming from the attacker’s ASV system: we reinforce privacy protection by maximizing ASV *ineffectiveness* while keeping VA *effectiveness*. We use adversarial noise to disrupt the generation of anonymized enrollment data by the attacker. The consequent mismatch between enrollment and trial data then results in a less effective attack, independently of the attacker ASV system. Our contributions are as follows: (i) we describe the first preemptive defense against VA re-identification attacks using adversarial perturbations; (ii) we show that the proposed technique reinforces the privacy protection of a given VA system against *any* attacker ASV system, often achieving the “perfect privacy” goal of 50% EER; (iii) we show that the improvements to privacy protection persist even in the case in which the attacker is aware of the adversarial perturbation and attempts to remove it with purification methods [15].

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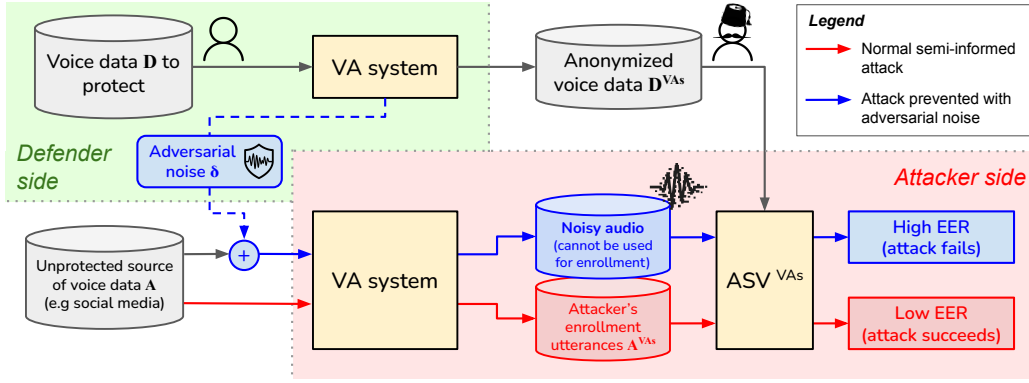


Fig. 1. Overview of the proposed preemptive voice anonymization technique.

II. RELATED WORK

A. Voice anonymization systems

State-of-the-art VA systems function similarly to Voice Conversion (VC) systems: in practice, VA is carried out by converting the voice of the original speaker to that of a different *pseudo-speaker*. In this paper, we use two baseline systems from the 2024 and 2026 editions of the VPC: B4 [16] and B5 [17]. We choose them as they were the most competitive in terms of privacy protection in the 2024 edition.

B4 performs anonymization by extracting *semantic tokens* from the utterance to anonymize with a HuBERT-based [18] semantic encoder. A Neural Audio Codec (NAC) [19] encoder extracts *acoustic tokens* from an utterance collected from the pseudo-speaker. Acoustic and semantic tokens are combined using an autoregressive transformer to generate a new set of acoustic tokens that encode the content of the original utterance, but the voice identity of the pseudo-speaker. These are then decoded to a waveform using the NAC decoder.

B5 extracts feature vectors from the input speech with a wav2vec 2.0 backbone [20] and discretizes them with a Vector Quantization (VQ) layer [21]. The result is fed into a HiFi-GAN [22] to synthesize the output waveform. Additionally, the HiFi-GAN is conditioned on a one-hot representation of the pseudo-speaker.

Both B4 and B5 operate upon discrete representations of the input utterance: semantic tokens in B4 and VQ codewords in B5. This shared structure enables the formulation of an optimization objective for adversarial noise generation that can be applied to both systems. Further details are provided in Section III.

B. Privacy evaluation via automatic speaker verification

As per the VPC evaluation protocol, the defender uses anonymization system VA_S to convert unprotected utterances D into the anonymized utterances D^{VA_S} . The attacker has access to unprotected reference utterances A belonging to the speakers they intend to re-identify within D^{VA_S} . To that end, they train an ASV system on a further dataset T . The VPC privacy evaluation protocol assumes a *semi-informed* attacker [4]: under this worst-case scenario, the attacker knows

which system VA_S was used by the defender to transform D into D^{VA_S} . They are able to use the same system to anonymize the enrollment utterances A and training data T , producing A^{VA_S} and T^{VA_S} , respectively. They use the latter to train a speaker verification system ASV^{VA_S} specifically adapted to VA_S . They then attempt the re-identification attack by using D^{VA_S} as trials and A^{VA_S} as enrollment utterances.

This approach constitutes the strongest attacks known to date [5], and is therefore considered to provide a lower bound for the privacy protection provided by a VA system (with some notable exceptions [14]). Preemptive anonymization is designed to hinder a semi-informed attack, and therefore raise this lower bound.

C. Adversarial perturbations in audio security and privacy

To the best of our knowledge, this is the first work to investigate adversarial perturbations as a means of enhancing the privacy protection of VA systems. The closest line of related research studies adversarial perturbations to defend against voice cloning [23]–[26]: these methods focus on generating noise that disrupts the output embeddings of one or more speaker encoders, preventing the use of protected utterances in a voice cloning system. Our approach is different in that we generate noise that degrades at the waveform level the reference enrollment utterances used by the attacker. Comparing them to the anonymized trials produces unreliable similarity scores and causes the re-identification attack to fail.

III. PREEMPTIVE DEFENSE AGAINST RE-IDENTIFICATION ATTACKS

The semi-informed attack scenario is based on two main assumptions: (i) the attacker somehow obtains unprotected enrollment utterances A of the speakers they intend to re-identify within the anonymized data D^{VA_S} ; and (ii) the attacker knows which system VA_S was used by the defender to anonymize D into D^{VA_S} . Assumption (i) requires the existence of a source from which the attacker can retrieve unprotected speech data of the speakers of interest. Social media provides a natural example, as users generally wish to remain recognizable and are therefore unlikely to anonymize their voices. The attacker could also gain unauthorized access to private speech data

that would not normally be anonymized, such as voice messages exchanged through messaging services. Our proactive anonymization method introduces a third assumption, (iii): the defender has some control over the speech data A that the attacker may potentially obtain. The defender can then protect these utterances in advance (e.g. before upload to social media) by adding adversarial perturbations, thereby preventing the attacker from using them as enrollment data. The proposed approach is illustrated in Figure 1.

The perturbations must respect two requisites. First, they should preserve the quality of the protected utterance; second, they cannot be optimized directly using ASV^{VA_S} since the defender cannot know the ASV system used by the attacker. However, the defender does know the VA system they used to generate D^{VA_S} . For a given utterance $\mathbf{u}$, the defender can then generate an adversarial perturbation δ in white-box fashion so that passing $\mathbf{u} + \delta$ through VA_S will result in a highly degraded output: in other words, they prevent the protected signal from being anonymized. If the attacker’s enrollment data A is preemptively protected with these perturbations, anonymizing it into A^{VA_S} will result in noisy data that will mislead ASV^{VA_S} , making the semi-informed attack fail. The attacker cannot circumvent the noise by using a different VA system: according to previous research, this would cause a distribution mismatch between A^{VA_S} and D^{VA_S} , resulting in a poor attack [14].

Exploiting the white-box setting, the adversarial perturbation is generated by targeting the internal modules of a given VA system. We describe this process first for B5, then for B4.

A. Attack on B5

Consider an utterance $\mathbf{u} \in \mathbb{R}^L$ to be anonymized. As an intermediate processing step, the utterance is encoded into a set of feature vectors using wav2vec 2.0 [20], with $\mathbf{h}_t$ being the feature vector at time frame t . Each $\mathbf{h}_t$ is subsequently mapped to a codeword by a VQ layer. Our goal is to find a perturbation $\delta \in \mathbb{R}^L$ which degrades the output of B5 as much as possible when added to $\mathbf{u}$. To this end, we maximize the number of *codeword flips*, defined as time frames for which the perturbed utterance is assigned a VQ codeword different to the original codeword. The resulting optimization objective is:

$$\max_{\delta} \frac{1}{T} \sum_{t=1}^T \mathbb{1} [\text{VQ}(\mathbf{h}_t) \neq \text{VQ}(\mathbf{h}_t^{\delta})] \quad (1)$$

where $\mathbb{1}[\cdot]$ is the indicator function, T is the total number of time frames, and $\mathbf{h}_t^{\delta}$ is the feature vector extracted from $\mathbf{u} + \delta$ at time step t . This objective cannot be optimized via gradient descent because of the non-differentiable VQ operation; however, an equivalent, differentiable surrogate objective can be formulated as:

$$\max_{\delta} \frac{1}{T} \sum_{t=1}^T \|\mathbf{h}_t - \mathbf{h}_t^{\delta}\|_1 \quad (2)$$

whereby we attempt to push $\mathbf{h}_t$ outside of its original Voronoi region [21] to produce a codeword flip. We found the L1

TABLE I
PRIVACY PRESERVATION RESULTS ON *libri_test*. EER (%) $\uparrow$ IS REPORTED FOR THE VPC 2024 AND VPC 2026 ATTACKERS.

System		VPC’24 attacker	VPC’26 attacker
B4	Baseline (no noise)	30.26	14.33
	+ δ noise, $\varepsilon = 0.05$	36.63	19.99
	+ δ noise, $\varepsilon = 0.1$	41.18	30.59
	+ δ noise, $\varepsilon = 0.3$	46.67	44.51
	+ δ noise, unbounded	47.51	45.14
B5	Baseline (no noise)	34.34	21.28
	+ δ noise, $\varepsilon = 0.05$	35.75	26.26
	+ δ noise, $\varepsilon = 0.1$	41.93	37.30
	+ δ noise, $\varepsilon = 0.3$	45.15	47.36
	+ δ noise, unbounded	46.33	46.64

distance to produce more consistent results than the more conventional mean squared error, since it produces smaller loss values and therefore more stable gradients.

Following other works [27], we limit the impact of perturbations upon audio quality by constraining δ to a ball of radius ε . This value can be tuned to adjust the privacy-utility tradeoff, with larger ε resulting in stronger preemptive protection but potentially greater degradation to quality.

In B5, if inference is done in batches, utterances are normally zero-padded to the right. Preliminary experiments showed that this padding can reduce the effectiveness of adversarial noise. To account for this, we also apply right zero-padding to $\mathbf{u}$ while optimizing (2), making δ robust to the padding introduced at inference time. The padding expands $\mathbf{u}$ to a different random length $L' > L$ at every iteration step. Adversarial noise is not applied to the padded area (δ is always of length L). Code of the optimization routine is available at https://github.com/eurecom-asp/preemptive_VA_defense.

B. Attack on B4

B4 includes a similar quantization step whereby the input utterance $\mathbf{u}$ is tokenized into semantic tokens using a semantic encoder. An optimization objective similar to (1) can therefore be applied by replacing the VQ layer with the tokenization operation. In practice, we again optimize (2) instead, where $\mathbf{h}_t$ are now the pre-tokenization feature vectors extracted using the semantic encoder. Since B4 does not support batched inference, no padding is ever applied to the inputs; therefore, δ does not need robustness to zero-padding.

Note that the surrogate objective (2) is generic enough to be applied to any VA system that extracts time-ordered embeddings from the input utterance.

IV. EXPERIMENTAL SETUP

A. Optimization parameters

We optimize δ for each utterance with Projected Gradient Descent (PGD) [28]. Gradients are computed with Adam [29]. Optimization is carried out until one of the following conditions are met: (a) we reach a maximum of 700 iterations;

and (b) the value of (1) is at least 0.95, implying that a codeword flip has occurred for 95% of the noisy feature vectors $\mathbf{h}_t$. The Adam learning rate is set to 10^{-4} for the first 300 iterations and 10^{-3} thereafter. The maximum duration of the random padding L' is set to 3 seconds. To explore the tradeoff between privacy protection and quality preservation, we perform experiments with ε values of 0.3, 0.1, 0.05. We also report results for unbounded noise, equivalent to $\varepsilon = \infty$.

B. Privacy preservation metrics

To show that preemptive anonymization works independently of the attacker ASV system, we perform privacy evaluation using the attackers of both the 2024 and 2026 VPC editions. The former is based on ECAPA-TDNN [30], while the latter additionally uses a WavLM front-end [31].

Privacy evaluation builds upon the standard VPC protocols described in [4], [5]. The privacy evaluation metric is the ASV^{VAS} EER. In our primary evaluation scenario, the defender applies adversarial perturbations to the enrollment partition of the *libri_test* dataset. Following the usual VPC-defined semi-informed attack, these protected utterances are further anonymized and used as enrollment data A^{VAS} . The corresponding ASV^{VAS} is trained using an anonymized version of *libri_train-360*.

Additionally, we consider a secondary privacy evaluation scenario involving a stronger attacker who is aware that the enrollment utterances have been protected with adversarial perturbations and who attempts to remove them through adversarial noise purification. We evaluate three purification methods. The first, proposed in [32], adds Gaussian noise to protected utterance; we consider standard deviations of 0.05 and 0.01. AudioPure [33] relies on an unconditional diffusion model trained using clean speech; we apply a single purification step. Finally, DAP [15] uses a diffusion model trained using speech data containing adversarial perturbations.

C. Quality preservation metrics

Following the VPC protocol, our primary quality preservation metric is the Word Error Rate (WER) of the same Automatic Speech Recognition (ASR) system used in the 2024 and 2026 VPC. Unlike the standard VPC setting, for which the WER is measured using anonymized trial utterances D^{VAS} , our objective is to preserve the intelligibility of enrollment utterances A after adversarial perturbation but before anonymization. We therefore compute the WER using perturbed enrollment utterances $A + \delta$ and compare it to the WER obtained by the same ASR system when applied to original enrollment utterances A ; the smaller the gap, the better the quality preservation.

WER only captures intelligibility from the perspective of an ASR system. For more comprehensive quality assessment, we also compute the following secondary quality metrics between original and noise-protected enrollment utterances: Signal-to-Noise Ratio (SNR) expressed in dB, Perceptual Evaluation of Speech Quality (PESQ) [34] and Short-Time Objective Intelligibility (STOI) [35].

TABLE II
QUALITY PRESERVATION RESULTS ON *libri_test_enrolls*. THE WER (%) OF THE ORIGINAL DATA IS 1.85.

System		WER ↓	SNR _{dB} ↑	PESQ ↑	STOI ↑
B4	+ δ noise, $\varepsilon = 0.05$	2.05	49.87	4.48	0.99
	+ δ noise, $\varepsilon = 0.1$	2.39	43.90	4.30	0.99
	+ δ noise, $\varepsilon = 0.3$	7.01	34.38	3.51	0.99
	+ δ noise, unbounded	29.58	26.70	2.55	0.97
B5	+ δ noise, $\varepsilon = 0.05$	1.88	49.88	4.49	0.99
	+ δ noise, $\varepsilon = 0.1$	1.97	43.89	4.30	0.99
	+ δ noise, $\varepsilon = 0.3$	2.19	34.39	3.53	0.99
	+ δ noise, unbounded	2.59	27.32	2.61	0.97

V. RESULTS

A. Privacy protection

Privacy results are reported in Table I. With the VPC 2024 attacker ASV system, B4 and B5 obtain baseline EERs of approximately 30% and 34%, respectively. Preemptive protection with unbounded adversarial perturbations ($\varepsilon = \infty$) increases EERs to 47% for B4 and 46% for B5, approaching the perfect privacy level of 50%. The VPC 2026 attacker is substantially stronger, and reduces the baseline EER to 14% for B4 and 21% for B5. Nevertheless, the proposed adversarial perturbations increase the EERs to approximately 45% and 46%, respectively, again bringing performance close to the perfect privacy level. These results suggest that proactive protection remains effective across different attacker ASV systems. This is because adversarial perturbation targets the VA system itself, rather than a specific ASV model. When objective (1) reaches a sufficiently high value, casual listening tests reveal that the anonymized enrollment utterance becomes largely unintelligible,¹ obscuring identifiable information regardless of the ASV system.

The method remains similarly effective when the perturbation magnitude is bounded by $\varepsilon = 0.3$. Across both B4 and B5, and for both the 2024 and 2026 attackers, the EERs again reach approximately 46%. Privacy gains drop when the perturbation bound is reduced to $\varepsilon = 0.1$. For the 2024 attacker, the EERs decrease to approximately 41% for both B4 and B5. The stronger 2026 attacker lowers them further, to 30% and 37%, respectively. Even so, these values remain approximately twice as high as the corresponding baseline EERs. Finally, a smaller bound of $\varepsilon = 0.05$ still provides modest privacy improvements.

B. Quality preservation

Quality preservation results are shown in Table II. In general, higher noise bounds ε lead to greater impacts on quality. The baseline WER for unprotected enrollment utterances is 1.85%. For B5, the addition of adversarial perturbation causes only a minor increase, even in the unbounded setting $\varepsilon = \infty$, for which the WER reaches 2.6%. Smaller perturbation bounds

¹Audio samples are provided in the supplementary material.

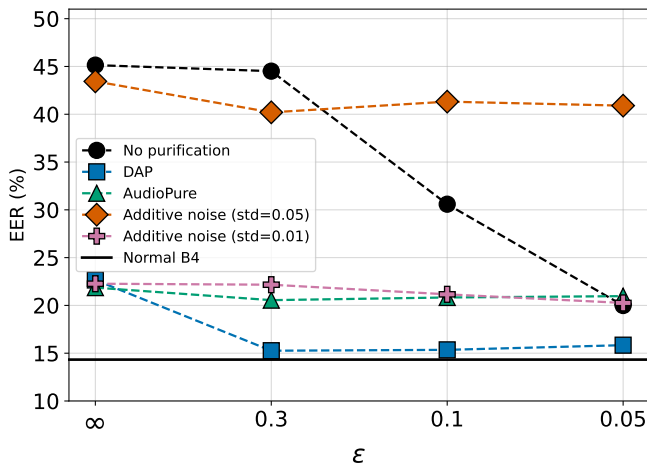


Fig. 2. Preemptive privacy protection results when the attacker attempts to remove the adversarial noise δ from the enrollment utterances with various purification methods. The experiments are run in the case of B4 with the VPC 2026 attacker (worst case scenario in terms of privacy protection).

ϵ have an even smaller effect. The secondary quality metrics show a similar trend. The SNR is 27.3 dB for the unbounded setting and increases as ϵ decreases, indicating that adversarial perturbations remain imperceptible relative to the speech signal. Intelligibility is fully preserved; all STOI scores are close to their maximum value of 1. PESQ scores also remain high, ranging from approximately 3.51 ($\epsilon = 0.3$) to 4.48 ($\epsilon = 0.05$) for a maximum value of 4.5.

In contrast, perturbations generated for B4 have a less negligible impact on the WER: for the unbounded setting, $\epsilon = \infty$, the WER rises unexpectedly to 29.6%. This degradation is substantially reduced when the perturbation is bounded, with $\epsilon = 0.3$ lowering the WER to approximately 7% and tighter bounds yielding further improvements. Despite the rise in WER, the secondary quality metrics are comparable to those obtained for B5, indicating that the perturbation is not larger in magnitude. Informal listening tests lead to the same conclusion. Further investigation suggests that the cause is likely an unintended *transfer* effect: although δ is optimized to disrupt B4, it may also interfere with the backbone of the specific ASR system used for evaluation. We find partial support for this hypothesis by repeating the experiment with a different ASR system² for which, in the unbounded setting, the WER is substantially lower at 6.6%.³

Overall, these results indicate that the quality of protected utterances is sufficiently preserved for use in other downstream tasks. In particular, the high PESQ and STOI scores suggest that δ has only a limited perceptual impact, making the method suitable for the protection of speech shared on social media (see use case described in Section III).

²<https://huggingface.co/speechbrain/asr-streaming-conformer-librispeech>

³The cause of this transfer remains unclear, since B4 and the original VPC ASR system do not share any internal modules. Interestingly, B5 is architecturally closer to the VPC ASR system; both use wav2vec-based backbones, yet no comparable transfer effect is observed.

C. Use of adversarial noise purification by the attacker

Figure 2 presents results for the scenario in which the attacker is aware of the presence of adversarial perturbations and attempts to remove them through purification. For conciseness, we consider the most challenging defender setting: using B4, which provides weaker privacy protection than B5, against the stronger VPC 2026 attacker.

Most purification methods are successful in attenuating adversarial perturbation, but also introduce distortions which have the effect of reducing the effectiveness of the attack. For instance, AudioPure and additive Gaussian noise with a standard deviation of 0.01 yield EERs of between 20% and 22% across all values of ϵ , remaining above the baseline EER of 14.3%. Gaussian noise with larger standard deviations degrades the signal even more than optimized perturbations δ , resulting in an even stronger attack disruption. DAP is the only method that almost completely removes the effect of proactive protection for $\epsilon \leq 0.3$.

One possible direction for future work would therefore be to make the adversarial perturbation more robust to purification, continuing the cat-and-mouse dynamic between defenders and attackers that characterizes the recent VA literature [36]. Even when purification is successful, however, proactive protection still increases the cost of carrying out an effective semi-informed attack. In the worst-case scenario, where the attacker is aware of the defense, they must still devote computational resources to training and applying a suitable purification method such as DAP, which requires the use of a computationally demanding diffusion model. The proposed approach therefore remains a meaningful deterrent by raising the effort required from the attacker.

VI. CONCLUSIONS

We presented the first approach to voice anonymization in which a defender acts preemptively to hinder potential re-identification attacks. Adversarial perturbations are used to protect clear speech data that might be obtained by an attacker, thereby making these utterances unsuitable for use as enrollment data in a semi-informed attack.

Our results show that the proposed method greatly enhances the privacy protection of a given voice anonymization system independently of the attacker ASV system. Speech quality is preserved, thereby safeguarding the use of treated enrollment data for other downstream tasks. Adversarial noise is applied to enrollment utterances retroactively, rather than to anonymized trial utterances; thus, the approach is complementary to regular voice anonymization, and enhances the protection of trial utterances that were anonymized even before the adversarial perturbations were added to the enrollment utterances.

Our results demonstrate the potential of preemptive anonymization. Future work should improve adversarial noise generation, either by increasing robustness to purification or by further reducing perceptual impacts, for instance through psychoacoustic masking [27], [37].

VII. AI-GENERATED CONTENT DISCLOSURE

LLM-based agents were used to assist in the development of some parts of the source code. LLM-based text generation was used to correct and rephrase part of the manuscript. The research question, proposed methodology and evaluation setup, as well as the manuscript structure, were formulated exclusively by the authors, without the assistance of AI.

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